MODULE 02 · SECTION 1 - FUNDAMENTALS OF GEN AI

AI, machine learning and generative AI: the vocabulary

The terms every other module assumes, and the near-misses the exam uses against you.

SECTION 1 of 4 EXAM WEIGHT ~30% MODULE 2 of 33

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On a definition question, the wrong options are rarely nonsense. They are **true of something else**.

Four boxes, one inside the next

01

Artificial intelligence

Machines that mimic human abilities: see, understand language, reason, decide.

02

Machine learning

AI that learns patterns from data instead of following written rules.

03

Deep learning

Machine learning with many-layered neural networks.

04

Generative AI

Deep learning models that create new content.

ON THE EXAM

Nested, not side by side. Generative AI is a kind of deep learning, which is a kind of machine learning, which is a kind of AI.

Rules written by people, or learned from data?

TRADITIONAL SOFTWARE

A person writes the rules

- If balance is over the limit, decline
- Predictable and fully auditable
- Breaks when cases fall outside the rules
- Every change needs a programmer

MACHINE LEARNING

The system learns the rules

- Finds patterns in thousands of examples
- Handles messy, varied real-world input
- Improves as it sees more data
- Only as good as the data it learned from

THE TIEBREAKER

Machine learning replaces explicit programming with learning from data. That phrase, learning from data, is the exam's anchor.

Predict an answer, or create something new?

PREDICTIVE AI

Outputs a label or a number

- Is this transaction fraud? Yes or no
- How many sofas will sell next month?
- Which segment is this customer in?
- The answer comes from a fixed set

GENERATIVE AI

Outputs new content

- Draft a product description
- Create an image for a campaign
- Summarise a long report
- The output did not exist before

THE TIEBREAKER

Look at the output. A category or a number is predictive. New text, images, audio, video or code is generative.

What generative AI can produce

OUTPUT	BUSINESS EXAMPLE	GOOGLE MODEL FAMILY
Text	Emails, summaries, product copy, reports	Gemini
Code	Suggest functions, write tests, explain code	Gemini
Images	Campaign visuals, product mock-ups	Gemini image models; Imagen (retired)
Video	Storyboards, short promotional clips	Veo
Audio and music	Voiceovers, jingles	Chirp, Lyria

ON THE EXAM

The guide's own list of business uses: text, image, code and video generation, data analysis, and personalisation.

NLP: making sense of human language

- **NLP** uses machine learning to reveal the structure and meaning of text.
- **Classic tasks:** sentiment, extracting names and dates, classifying documents, translation.
- **Understanding** reads language; **generation** writes it. Both sit under NLP.
- **Google's Natural Language API** does sentiment, entity and content classification out of the box.
- **Not all NLP is generative.** Tagging a complaint as negative creates nothing new.

> THE TIEBREAKER
>
> Language in the question does not make it a generative AI question. Check whether the output is a label or new text.

A foundation model is trained once on a massive amount of data, then **adapted to a wide range of tasks**.

What makes a model a foundation model

- **Broad training.** Huge, varied data, not one labelled dataset for one task.
- **General purpose.** Summarise, translate, classify, draft, answer, reason.
- **Adaptable.** Steer it with prompts, ground it in your data, or tune it.
- **Costly to build.** Most organisations use one rather than train one.
- **Examples:** Gemini, Gemma, Imagen and Veo; Claude and Llama from other firms.

> ON THE EXAM
>
> Training your own foundation model from scratch is almost never the right exam answer for a business.

LLM or foundation model?

LARGE LANGUAGE MODEL

Trained mainly on text and code

- Reads and writes language
- Chat, summaries, drafting, code
- A type of foundation model
- Gemini and Gemma are examples

OTHER FOUNDATION MODELS

Trained on other kinds of data

- Images, video, audio, or a mix
- Imagen for images, Veo for video
- Also foundation models
- But not language models

THE TIEBREAKER

All LLMs are foundation models, but not all foundation models are LLMs.

Multimodal: more than one kind of input or output

- **Modality** means a type of data: text, image, audio, video, code.
- **A multimodal model** is trained on several modalities and can combine them.
- **Gemini** was built multimodal: read a photo, a chart, a recording or a video.
- **Business value:** one model for mixed inputs, such as a photo plus a note.
- **Text-only models** cannot see the photo, however long their context.

> ON THE EXAM
>
> When a question mixes inputs, such as images with text or audio, look for the word multimodal in the right answer.

Definitions that belong to the neighbours

IF AN OPTION SAYS	IT IS DESCRIBING
Fits a line to observed data to estimate a value	Regression, a predictive technique
Inspired by the brain; layers of connected nodes	A neural network, the engine of deep learning
Learns and improves from experience without explicit rules	Machine learning in general
Creates new content such as text, images, code or music	Generative AI
Trained on broad data and adapted to many tasks	A foundation model

THE TIEBREAKER

Before choosing, name the term each option really describes. Usually only one matches the term in the question.

Which of Wrenfield's ideas are generative?

WRENFIELD IDEA	WHAT IT OUTPUTS	KIND OF AI
Forecast sofa demand by store	A number	Predictive ML
Flag risky Wrenfield Pay transactions	Yes or no	Predictive ML
Tag chats as complaint, query or praise	A label	NLP classification
Draft product descriptions from specs	New text	Generative AI
Find products from a shopper's photo	Matching products	Multimodal search

THE TAKEAWAY

Only one of these five creates new content. Gen AI is one tool in the box, not the whole box.

> Wrenfield's board pack lists four AI projects for next year. The chief financial officer wants to know which one is generative AI, because it needs a separate approval.
>
> Each project is described by what it will produce.

Which project is generative AI?

A Forecast next month's sofa sales from three years of history

B Flag card transactions that are likely to be fraudulent

C Group customers into segments by purchase behaviour

D Write first drafts of product descriptions from specifications

◆ PAUSE AND DECIDE

Which project is generative AI?

A Forecast next month's sofa sales from three years of history

Outputs a number. Predictive machine learning.

B Flag card transactions that are likely to be fraudulent

Outputs yes or no. Predictive classification.

C Group customers into segments by purchase behaviour

Outputs groups. Machine learning, but nothing new is created.

D Write first drafts of product descriptions from specifications

Outputs new text that did not exist before.

THE SIGNAL

Judge by the output. New content is generative; a number, a label or a group is not.

> In a strategy meeting, a board member says an LLM and a foundation model are the same thing. Another says foundation models are only useful after they have been fine-tuned.
>
> The chief executive asks you to settle it with one accurate statement.

Which statement is accurate?

A. Every foundation model is a large language model

B. Every LLM is a foundation model, but foundation models also cover images, video and audio

C. LLMs are small models trained from scratch for a single task

D. A foundation model must be fine-tuned before it can be used

◆ PAUSE AND DECIDE

Which statement is accurate?

A Every foundation model is a large language model

Backwards. Imagen and Veo are foundation models but not LLMs.

B Every LLM is a foundation model, but foundation models also cover images, video and audio

Exactly Google's framing of the relationship.

C LLMs are small models trained from scratch for a single task

LLMs are large, broadly trained and general purpose.

D A foundation model must be fine-tuned before it can be used

Prompting alone is enough for many tasks. Tuning is optional.

THE SIGNAL

All LLMs are foundation models; not all foundation models are LLMs.

> A home insurer's claims team receives three things for every storm claim: photos of the damage, a written description, and a voicemail from the customer.
>
> They want one model to take in all three and draft a claim summary for an assessor to review.

What should the insurer look for in a model?

A. A multimodal foundation model such as Gemini

B. A text-only LLM with a very long context window

C. A speech-to-text service on its own

D. An image classifier trained on past damage photos

◆ PAUSE AND DECIDE

What should the insurer look for in a model?

A A multimodal foundation model such as Gemini

Reads images, text and audio together, then writes the summary.

B A text-only LLM with a very long context window

Context length does not let it see photos or hear audio.

C A speech-to-text service on its own

Handles the voicemail only, and writes no summary.

D An image classifier trained on past damage photos

Outputs a label for the photo. It ignores the rest and creates nothing.

THE SIGNAL

Mixed inputs in one task: photos, text and audio. That is the definition of a multimodal job.

> Wrenfield's contact centre wants every incoming chat tagged as a complaint, a query or praise, so it can be routed to the right team.
>
> A team leader calls this a generative AI project and asks for a generative AI budget.

Which description of the task is most accurate?

A It is generative AI, because it works with language

B It needs a diffusion model to produce the tags

C It is NLP classification: the output is a label, not new content

D It requires training a new foundation model from scratch

◆ PAUSE AND DECIDE

Which description of the task is most accurate?

A It is generative AI, because it works with language

Language is not the test. The output is what counts.

B It needs a diffusion model to produce the tags

Diffusion models generate images and video, not labels.

C It is NLP classification: the output is a label, not new content

Three fixed labels for each chat. Classic NLP classification.

D It requires training a new foundation model from scratch

Far too costly, and existing tools already classify text.

THE SIGNAL

Three fixed categories as the output means classification, even when the input is language.

Module 2 in five lines

- AI contains machine learning, which contains deep learning, which contains generative AI.
- Machine learning learns from data instead of following explicit rules.
- Predictive AI outputs a label or number; generative AI outputs new content.
- A foundation model is broadly trained and adapted to many tasks. All LLMs are foundation models.
- Language is not the test, and multimodal means mixed inputs. Judge by the output.